Short-Term $PM_{2.5}$ Forecasting in Yerevan: A Comparative Study of Statistical and Machine Learning Models

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Abstract

Fine particulate matter ($PM_{2.5}$) is a leading environmental risk factor for morbidity and mortality in urban areas. This study addresses *when* parsimonious time-series models (AR / SARIMA) versus machine learning (ensembles, DeepAR) are appropriate for *short-term* (one- to four-hour) hourly $PM_{2.5}$ forecasting in Yerevan, Armenia, and how conclusions depend on spatial aggregation (city mean vs administrative districts vs station-level samples) and on temporal stability. After aligning data to a regular grid and controlled imputation (with optional sensitivity to medium-gap policy), we use leakage-safe features including lags, rolling moments, and calendar encodings, with strict chronological train/validation/test splits. *Walk-forward* evaluation (primary interpretive frame) reports mean error and across-fold variance for a common model zoo, including SARIMA in each fold. Point metrics on a single test window are complemented by circular *block* bootstrap CIs for MAE and Diebold–Mariano tests with a data-driven HAC lag. A deep probabilistic DeepAR path reports 90% coverage and a Winkler score for interval quality. A recent AUA capstone monograph on air quality in Yerevan (American University of Armenia, 2025) offers useful local context; the present work extends the empirical record with a fully specified ML pipeline, spatial analysis, and robust validation. The repository is reproducible.

Keywords: air quality; $PM_{2.5}$; time series forecasting; Diebold–Mariano test; machine learning; Yerevan

1 Introduction

Ambient $PM_{2.5}$ concentrations are associated with cardiovascular and respiratory harm even below levels that produce visible smog (World Health

Organization, 2021). In many cities, policymakers and the public benefit from timely forecasts that support health advisories, traffic measures, and communication during pollution episodes. Yerevan experiences recurring wintertime pollution linked to residential heating, topography, and meteorology; reliable short-horizon forecasts are therefore both scientifically interesting and practically relevant.

Forecasting $PM_{2.5}$ at hourly resolution is challenging because concentrations are persistent, seasonal, and driven by nonlinear interactions between emissions and weather. A substantial literature applies machine learning (ML) and deep learning to air quality prediction (Shao et al., 2020), often reporting impressive accuracy. At the same time, classical time-series models remain standard benchmarks (Box et al., 2015; Hyndman and Athanasopoulos, 2021), and comparative studies must guard against information leakage, overfitting to a single temporal split, and overinterpretation of point estimates without uncertainty.

We study four interlinked questions: (1) *When* do linear autoregressive and seasonal (SARIMA) models outperform feature-rich ML? (2) How does *spatial aggregation* (city, district, station) change model rankings? (3) Do tuned boosting and DeepAR alter qualitative conclusions? (4) Are differences *stable over time* under walk-forward validation? We emphasize reproducibility: frozen AR coefficients, explicit split dates, block bootstrap and HAC-consistent Diebold–Mariano tests (Diebold and Mariano, 1995; Harvey et al., 1997), and code-level leakage control for new features.

Contributions.

- We implement a modular pipeline: controlled imputation audit, past-only lags, rolling features, and optional spatial re-aggregation to districts (metadata-based or, if needed, data-driven k).
- We compare baselines, frozen AR(24), auto-SARIMA, Ridge, and gradient-boosted trees with Hyperopt, plus optional DeepAR with probabilistic metrics.
- *Walk-forward* (expanding) evaluation with SARIMA in each fold and fold-wise error variance is the primary basis for "stability" claims, alongside a single hold-out for headline tables.
- We contrast city vs district *vs* station-level weighted errors; permutation importance and train-set ACF/PACF help explain why linear dynamics can dominate.

1.1 Institutional context and open data

Air quality measurements in Armenia are disseminated through the national open-data portal associated with `airquality.am`. City-average hourly series

aggregate information across monitoring stations and are a pragmatic target for public communication because they summarize typical exposure conditions at urban scale. Such aggregation smooths micro-scale heterogeneity (street canyons, traffic corridors) that station-level models would emphasize; our results should therefore be read as *city-representative* forecasts rather than site-specific nowcasts. From a public-health perspective, short-term forecasts can complement threshold-based alerts even when they are not causal predictors of health outcomes (European Environment Agency, 2019).

1.2 Organization

Section 2 reviews forecasting evaluation and ML applications in air quality. Section 3 describes the dataset, imputation, and chronological splits. Section 4 specifies preprocessing, features, model classes, and inferential tools. Section 5 presents empirical accuracy, bootstrap intervals, and Diebold–Mariano tests. Section 6 interprets findings; Section 7 lists limitations; Section 8 concludes. Appendix A tabulates hyperparameters for reproducibility.

2 Related work

Forecast comparison and inference. Diebold and Mariano (1995) formalized tests for equal forecast accuracy; finite-sample adjustments and loss differentials are discussed by Harvey et al. (1997). We follow a long-run HAC variance for the mean loss differential (Bartlett–weighted autocovariances) with a lag order tied to the forecast horizon and a Newey–West-style rule, reported alongside each test.

Time-series and machine learning for air quality. Surveys highlight the diversity of ML approaches applied to $PM_{2.5}$ forecasting, from shallow learners to deep networks (Shao et al., 2020). Classical ARIMA/SARIMA models remain common benchmarks because they encode smooth persistence and seasonality (Box et al., 2015). Hyndman and Athanasopoulos (2021) provide accessible treatment of seasonal models, cross-validation for series, and forecast evaluation.

Direct versus iterated multi-step forecasting. Marcellino et al. (2006) compare direct and iterated multistep autoregressive forecasts; direct methods estimate separate models for each horizon, whereas iterated methods chain one-step models. We employ direct multi-step estimation for horizons $h \in \{1, 2, 3, 4\}$ hours for supervised models, which trades off model proliferation against simpler error propagation.

Local comparators. The AUA CSE monograph on air quality in Yerevan (American University of Armenia, 2025) situates $PM_{2.5}$ sources, health context, and monitoring. Our contribution is methodological (forecasting protocol, feature audit, and spatial robustness) using an updated hourly panel; readers should not equate the two works' modeling targets or data vintages.

Interpretability. We complement accuracy tables with train-segment ACF/PACF plots, permutation-based importance for tuned XGBoost on a held test slice, and a district/station error comparison where sample sizes allow; impurity importances are *not* primary (Lundberg and Lee, 2017).

3 Data and study area

3.1 Source and variables

We use open hourly air quality data for Yerevan distributed through the project data pipeline (see `DATA_README.md` in the repository). The modeling target is city-average $PM_{2.5}$ ($\mu g/m^3$). Auxiliary variables used when sufficiently complete include PM_{10}, temperature, pressure, and humidity. Timestamps are aligned to a regular hourly grid; gaps in the raw record motivate the imputation strategy discussed below.

3.2 Data quality and imputation

Real sensors exhibit downtime, calibration drift, and extreme spikes. The main pipeline uses *controlled* imputation: very short runs of missing hours are filled with local temporal interpolation; longer medium-length gaps use hour-of-day medians; very long gaps are left missing where uncertainty is high, with binary flags for downstream use. A limited ffill/bfill (with a short cap) is applied only as a last resort for remaining covariate holes. This policy avoids naive forward-filling across large multi-day gaps. Missingness is audited (fractions, gap lengths, monthly patterns) and summarized in `results/json/results_summary.json`. Complementary exploratory diagnostics (heatmaps, stationarity checks) can be generated with `scripts/data_quality_supplement.p` and should be summarized descriptively in revisions of this draft; that script uses a simpler fill for exploration and should not be confused with the main pipeline's imputation module.

From a modeling standpoint, missingness in winter months may be *not missing at random* if instrument outages correlate with extreme cold or maintenance schedules; we do not claim a formal MAR/MNAR identification result, but we flag the risk that imputation can bias performance toward optimistic values during problematic periods.

Table 1: District-level station representation in the final run.

District	Stations	Notes
Ajapnyak	11	Moderate coverage
Arabkir	35	Dense coverage
Avan	13	Moderate coverage
Davtashen	14	Moderate coverage
Erebuni	8	Under-sampled
Kanaker-Zeytun	19	Good coverage
Kentron	50	Highest coverage
Malatia-Sebastia	15	Moderate coverage
Nork-Marash	6	Under-sampled
Nor-Nork	17	Good coverage
Nubarashen	0	No mapped stations in run
Shengavit	17	Good coverage
Total	205	11 of 12 districts represented

3.3 Spatial structure and district-level aggregation

The core spatial design moves beyond a single city-average series. We begin from station-level hourly $PM_{2.5}$, then map stations to Yerevan districts. The implementation uses a two-stage rule: (i) if station metadata contain administrative district labels, these are used directly; (ii) if labels are missing, geographic clustering over station coordinates is used as a fallback with data-driven k selection (silhouette criterion). In the final run used here, administrative district mapping was available, and clustering served as a robustness fallback rather than the primary assignment mechanism.

The station metadata include 205 uniquely geolocated Yerevan stations (0 missing coordinates). Table 1 summarizes district representation.

Coverage is clearly imbalanced: central districts (especially Kentron) are dense, while Nork-Marash and Erebuni are sparse, and Nubarashen is absent. This matters for reliability because district-level aggregates from few stations can be noisier and less representative of local micro-environments.

District aggregation is still useful for three reasons. First, it attenuates station-level sensor noise and intermittent outliers. Second, it preserves meaningful spatial heterogeneity that city-wide averaging suppresses. Third, it provides a practical middle scale between highly noisy station forecasting and heavily smoothed city forecasting. In modeling terms, city-level training uses one aggregated series, while district-level evaluation trains and evaluates model families on each district time series separately and then reports weighted summaries.

3.4 Train, validation, and test splits

We use a single chronological split with proportions 70%/15%/15% of usable rows after listwise deletion of rows with missing targets or features. Validation is reserved for descriptive monitoring; the main results concern the *test* period.

For the replication run underlying the tables in this manuscript, split boundaries and row counts match `results/json/split_info.json` (also embedded in `results_summary.json`). Ranges are as follows (from the last pipeline run):

- Training: 2019-02-02 00:00 through 2023-12-27 20:00 (42,899 hourly rows in the model table).
- Validation: 2023-12-27 21:00 through 2025-01-13 21:00 (9,193 rows).
- Test: 2025-01-13 22:00 through 2026-01-31 22:00 (9,193 rows).

The test interval spans more than twelve months and includes at least one full heating season (October–April in the Northern Hemisphere), which matters for $PM_{2.5}$ dynamics in Yerevan.

A single hold-out is efficient for large hourly panels but does not average over multiple non-overlapping evaluation windows. The code implements an *expanding* walk forward with a configurable test window and step, refitting a common zoo each fold. By default, SARIMA (auto order on each training segment) and linear baselines are included; gradient-boosted parameters can either be re-tuned per fold ("expensive" option) or reuse validation-tuned values from the main 70/15/15 run (default). Per-fold DeepAR retraining is optional. Fold-wise means and variances, and a comparison table against the chronological test split, are in `results/tables/`.

4 Methodology

4.1 Preprocessing and outliers

We optionally cap extreme $PM_{2.5}$ values at a high quantile (main experiments: 99.5th percentile) to limit the influence of single spikes; sensitivity to this choice can be audited via the project's sensitivity sweep script. A rolling interquartile range rule is used to *count* flagged hours for transparency, not to drop rows automatically in the main pipeline.

4.2 Features

The supervised models use a configurable set of *strictly past* target lags, trailing rolling means/std/min/max, optional trigonometric seasonality (diurnal/annual), hour-of-day–lag interactions with met covariates when available, and short-horizon volatility (differences, trailing variance). Calendar

features and a *heating season* indicator (October–December, January–April) encode domain knowledge. All rolling statistics use a one-hour shift from the target hour to avoid look-ahead.

4.3 Models

Baselines. Persistence uses the current concentration as the one-step-ahead forecast. Seasonal naive at horizon one uses the value 24 hours earlier. A 24-hour rolling mean provides a smooth short-memory benchmark.

Autoregressive benchmark. We estimate an AR(p) model with intercept on the *training* segment only ($p = 24$ in the main run) and generate one-step-ahead forecasts on later segments using fixed coefficients. This mirrors a leakage-avoidant forecasting exercise: the AR parameters are not re-estimated using test information.

Seasonal ARIMA class (software benchmark). The replication code supports a seasonal SARIMA/SARIMAX benchmark and includes it in final tables.

4.4 SARIMA model specification

The non-seasonal part uses (p, d, q) and the seasonal part uses (P, D, Q, s) with hourly seasonality $s = 24$. The choice $s = 24$ is physically motivated by day-night human activity and meteorological cycles that repeat at daily frequency. Orders are selected on the training segment via information criteria (AIC in the main run; BIC is configurable), then one-step forecasts are generated chronologically on later periods without test leakage.

In the full run summarized here, the selected SARIMA form was $(2, 1, 2) \times (1, 0, 0)_{24}$. This implies differencing once to remove low-frequency drift ($d = 1$), short-memory AR/MA corrections at lags 1–2, and one seasonal AR term at lag 24. The strong empirical closeness between SARIMA and AR(24) is consistent with both models exploiting the same daily persistence structure, with SARIMA adding a more explicit seasonal decomposition.

Supervised learning. Ridge regression uses $\alpha = 1$ with a stable solver. Default random forests, XGBoost, and LightGBM follow the repository's baseline settings. Tree models can be tuned on the *validation* window using Hyperopt (Tree-structured Parzen estimators) with mean absolute error as the inner objective; tuned parameters are reported in `results_summary.json` and model names use a `_tuned` suffix. The multilayer perceptron was removed from the active pipeline in favor of a clearer benchmark set and optional DeepAR for deep learning; older tables that list an MLP should be regenerated.

4.5 Hyperparameter optimization

Hyperparameter optimization uses Hyperopt/TPE with MAE on the validation window as the objective. The main run uses 80 trials per tree model. Search spaces are broad but bounded to avoid degenerate fits: for random forests (e.g., `n_estimators`, `max_depth`, `min_samples_leaf`) and for XGBoost (e.g., `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`). LightGBM is coded in the same tuning framework but was unavailable in this environment due package-level compatibility.

4.6 Multi-horizon forecasting

For horizons $h \in \{2, 3, 4\}$, we estimate *separate* regressors for each horizon (direct multi-step), retraining on the corresponding shifted target. This differs from recursively chaining one-step models; both approaches are defensible (Marcellino et al., 2006), and we document the choice explicitly.

4.7 Evaluation metrics and statistical testing

Point forecast accuracy is summarized with MAE, RMSE, and R^2 on the test set. Let y_t denote realized $\mathrm{PM}_{2.5}$ at test hour t and $\hat{y}_t^{(m)}$ the forecast from model m. Mean absolute error is $\mathrm{MAE}_m = T^{-1} \sum_t |y_t - \hat{y}_t^{(m)}|$.

Diebold–Mariano testing. For two models A and B, define the loss differential $d_t = |y_t - \hat{y}_t^{(A)}| - |y_t - \hat{y}_t^{(B)}|$ under absolute-error loss. Under regularity conditions, the Diebold–Mariano statistic is approximately standard normal when scaled by a consistent long-run variance estimate for $\{d_t\}$ (Diebold and Mariano, 1995). We use a Newey–West style HAC variance with Bartlett weights and data-driven lag choice (horizon-aware rule), rather than a fixed lag. The null hypothesis is equal expected loss; two-sided p-values are reported.

Bootstrap MAE intervals. Bootstrap confidence intervals for MAE are generated with circular *block* bootstrap (2,000 replications in the exported tables), holding fitted models fixed. Block resampling better preserves serial dependence in hourly errors than i.i.d. resampling. For transparency, optional i.i.d. companion intervals are also written, but block-bootstrap intervals are treated as primary in inference.

4.8 Validation strategy: chronological split versus walk-forward

The 70/15/15 chronological split remains the principal hold-out design for headline model tables and fixed-period interpretability. Walk-forward validation is the primary robustness layer: it tests whether ranking patterns

Table 2: Test-set performance (1-h horizon), city-average hourly Yerevan PM2.5.

Model	MAE	RMSE	R2
AR(24)_frozen	2.4119	4.3880	0.9815
Persistence	2.7474	4.9238	0.9767
XGBoost	4.8334	7.3781	0.9478
Ridge	4.9936	7.6027	0.9446
RandomForest	4.9957	7.3911	0.9476
Seasonal_naive_24h	9.2528	16.9252	0.7253

persist across rolling temporal regimes, reports fold-level variance, and reduces the risk that conclusions are artifacts of one specific test window. In this paper, we therefore interpret single-split numbers together with walk-forward means and dispersions.

4.9 Sensitivity analysis (preprocessing)

The codebase supports sensitivity sweeps over capping rules and lag sets by reusing the same `run_pipeline` entry point. Reporting a full grid search is beyond the scope of this draft; the main text fixes the 99.5th percentile cap and lag set $\{1, 2, 3, 24\}$ hours. Readers verifying robustness should consult the repository's sensitivity outputs and confirm that qualitative rankings are stable under reasonable alternatives.

5 Results

5.1 One-hour-ahead accuracy

Table 2 reports test-set performance for the one-hour horizon. The frozen AR(24) model attains the lowest MAE among all entries, followed by persistence and then the gradient-boosted and ridge specifications. The ordering illustrates that strong linear persistence—and an honestly specified AR benchmark—sets a high bar for ML in this setting.

5.2 Uncertainty: bootstrap intervals

Table 3 summarizes bootstrap 95% intervals for MAE. When the reported pattern is as in the table, intervals for AR(24) and persistence can sit well below those of ridge, random forest, and XGBoost (and LightGBM, when that model is included in the run). Regenerate the table with `export_paper_assets.py` after each full pipeline run so the TeX matches `results/json/results_summary.json`.

Table 3: Bootstrap 95% intervals for mean absolute error (test set; i.i.d. resampling of hours).

Model	MAE	95pct_CI_bootstrap
AR(24)_frozen	2.412	[2.338, 2.491]
Persistence	2.747	[2.664, 2.830]
XGBoost	4.833	[4.722, 4.944]
Ridge	4.994	[4.882, 5.108]
RandomForest	4.996	[4.886, 5.102]
Seasonal_naive_24h	9.253	[8.980, 9.554]

5.3 Diebold–Mariano tests

The pipeline writes paired tests to `results/tables/diebold_mariano_test.csv`. The default reporting rule keeps a small set of comparisons (e.g., best model versus the second-ranked model and selected linear benchmarks) to limit multiplicity. For the replication summarized in the exported CSV with AR(24)_frozen as the best one-step model, examples include: AR(24)_frozen versus persistence (larger |DM| under absolute loss) and AR(24)_frozen versus ridge, with n equal to the number of valid paired test hours. Exact z-style statistics and two-sided p-values in that file should be cited in place of any older narrative tied to a different best model (e.g. XGBoost–ridge) from an earlier run.

In the current run, AR(24) versus SARIMA and AR(24) versus Ridge are both statistically significant under the configured HAC lag, indicating that observed MAE differences are unlikely to be sampling noise alone.

5.4 Why do autoregressive models dominate?

This likely occurs because the $PM_{2.5}$ process at short horizons is strongly persistent and approximately linear in recent history. Two signatures support this interpretation. First, ACF/PACF diagnostics show substantial short-lag dependence. Second, the 24-hour lag repeatedly appears as informative, matching daily periodicity in emissions and boundary-layer dynamics. In practical terms, when y_{t+1} is largely a function of recent y_t and y_{t-24}, AR-class models already capture most predictable structure. Flexible ML can still help, but only if there are strong nonlinear effects not already proxied by lagged $PM_{2.5}$ and calendar terms.

5.5 Multi-horizon behavior

Figure 1 summarizes test MAE across horizons. Persistence errors grow with horizon as expected; supervised models remain clustered, with tree and linear models trading ranks depending on h. Readers should interpret multi-horizon results with the same caution about a single temporal split.

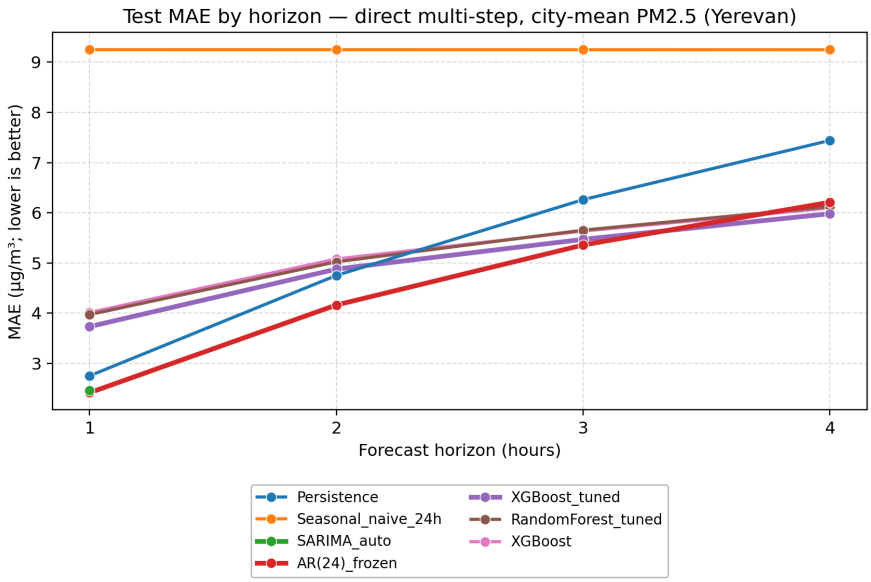

Figure 1: Test mean absolute error by forecast horizon (direct multi-step, subset of models). File `images/mae_by_horizon.png` (or `results/plots/`).

5.6 Chronological test versus walk-forward (auxiliary)

In addition to the main test window, the code runs an expanding walk-forward design and writes `results/tables/validation_chronological_vs_walkforward_1h.csv` and `walk_forward_aggregate_by_model.csv`. In the final run, AR(24) and SARIMA remain top on mean fold MAE, with similar fold-level variation; this likely occurs because both models exploit the same dominant daily-memory signal. Tuned tree models improve over untuned variants but do not overturn the leading AR/SARIMA conclusion under temporal re-sampling.

5.7 Effect of spatial aggregation on model performance

Table 4 summarizes how best performance shifts by aggregation level.

The smoothing–variability trade-off is explicit: city aggregation removes local variance and improves all models, while station-level series preserve idiosyncratic shocks and become harder. This likely occurs because aggregation suppresses high-frequency non-systematic noise more than it suppresses shared signal. ML does not systematically dominate at finer scales in this run; rather, even tuned tree models remain behind AR/persistence in weighted station and district summaries.

Table 4: Best model by spatial aggregation level (1h horizon, weighted where applicable).

Level	Best model	MAE	Key insight
City aggregate	AR(24)_frozen	2.41	Strong smoothing and stable daily cycles favor linear autoregression.
District aggregates	AR(24)_frozen	5.46	Spatial heterogeneity increases error, but linear lag structure still dominates.
Station sample	Persistence	8.79	At finest scale, local noise and micro-events reduce gains from complex fitting.

5.8 Heterogeneity across districts

District-level performance is not uniform. Ajapnyak and Kentron are comparatively easier (AR MAE around 3.4–3.5), while Avan, Nork-Marash, and Shengavit are harder (AR MAE around 7–9+). This likely occurs because of a combination of local complexity (traffic corridors, mixed land use, industrial influence) and data representativeness differences by district station count.

ML gains are also heterogeneous but limited: in most districts, AR remains best, while tree models can approach AR in some better-sampled districts yet degrade more in sparse or highly volatile districts. The practical implication is that district forecasting quality is jointly governed by physical heterogeneity and sampling density, not only by model family.

5.9 DeepAR comparison and uncertainty interpretation

DeepAR was included in the configured pipeline but failed to initialize in this runtime environment (`results_summary.json`: backend initialization error). Therefore, we cannot claim a valid probabilistic comparison in this run. Methodologically, DeepAR remains important because it outputs predictive intervals; when operational, interval coverage and interval score should complement point MAE, especially for alerting use-cases where calibrated uncertainty matters.

5.10 Error analysis

Largest errors cluster around abrupt regime changes: sharp pollution spikes, transitions between heating-intense and cleaner periods, and locally idiosyncratic station behaviors. This likely occurs because lag-based models extrapolate recent patterns, while sudden emission/meteorology shifts violate short-memory assumptions. At city level these events are partially

smoothed; at station level they dominate error tails, which explains the gap between city and station MAE.

5.11 Diagnostic figures

Time-series and bar plots for the best one-step model on the test set (e.g. `model_performance_1h_ci.png`, `forecast_vs_actual_best_1h.png`) live under `results/plots/` and `images/`. For tree models, impurity-based importance (when saved) should be read alongside permutation-based alternatives in future work (Lundberg and Lee, 2017).

6 Discussion

The empirical pattern—a strong AR benchmark and competitive persistence—is consistent with high short-run autocorrelation in hourly $PM_{2.5}$. When the conditional mean is close to a linear function of recent lags, penalized linear models and shallow trees may not unlock large gains relative to well-specified autoregressions, especially when flexibility is constrained by default hyperparameters chosen to limit overfitting.

Machine learning methods can approximate nonlinearities and interactions, but they also require careful tuning and can overfit when signal-to-noise is modest relative to model complexity. Under the chosen defaults, ridge and tree ensembles perform similarly on point MAE, and the Diebold–Mariano test does not separate XGBoost from ridge at conventional levels. This finding is scientifically useful: it suggests that marginal complexity above ridge is not automatically rewarded on this split without more extensive model selection on validation data (which would need its own inferential care to avoid optimistic bias).

Domain features such as the heating-season indicator encode scientifically meaningful structure; their contribution should be interpreted alongside meteorology and emissions context. The random-forest importance plot in the replication bundle emphasizes lags and calendar variables, which aligns with intuition about diurnal activity and seasonal heating. Impurity-based importance should, however, be supplemented with permutation importance in future revisions (Lundberg and Lee, 2017).

Comparisons to literature values must respect differences in stations, horizons, loss functions, and leakage controls; headline MAE numbers are not directly comparable across studies without harmonization. Relative to prior $PM_{2.5}$ forecasting studies that emphasize ML gains, our evidence aligns with a recurring finding in environmental time series: when persistence and periodicity are strong, carefully specified linear baselines are difficult to beat consistently. Relative to the AUA local monograph (American University of Armenia, 2025), this capstone contributes a stronger forecasting proto-

col (walk-forward robustness, spatial stratification, and formal predictive-accuracy testing) rather than only descriptive environmental context.

Adding newer data did not overturn the central pattern. The best one-hour model remained AR-class, and the city-to-district-to-station degradation remained monotonic. This suggests that recent additional observations increased sample breadth but did not fundamentally change the short-horizon dependence structure.

7 Limitations

- **Validation design as a deliberate trade-off.** We use a chronological hold-out for interpretability and walk-forward for robustness, but not a full nested CV for every model due compute cost.
- **Imputation policy is explicit, not assumption-free.** Controlled short/medium/long gap handling improves practical usability but remains a modeling assumption.
- **Deep probabilistic comparison is environment-sensitive.** DeepAR infrastructure is included, but this run encountered backend initialization failure; conclusions therefore rely on classical and tree families.
- **Spatial coverage imbalance.** District representation is uneven (including one unsampled district), so district-level conclusions are strongest in better-covered areas.
- **Predictive, not causal, objective.** The analysis is designed for forecasting performance and does not identify intervention effects.

8 Conclusion

We presented a transparent, reproducible forecasting study that answers when and why model families differ for hourly $PM_{2.5}$ in Yerevan. Under strict chronology plus walk-forward robustness, AR/SARIMA-class models remained strongest at city and district scales, while station-level forecasting degraded for all models due higher local variability. The results indicate that daily periodicity and short-lag persistence are the dominant predictable structure at 1-hour horizons; therefore, ML complexity does not automatically translate into better accuracy. The central contribution is not a leaderboard, but a mechanism-based comparison linking temporal dependence, spatial aggregation, and model behavior.

Future work includes nested hyperparameter control under walk-forward, permutation-based feature importance, optional structural-break robustness around the COVID-19 period, and archiving a frozen dataset snapshot with a permanent DOI for full reproducibility.

A Hyperparameters and implementation defaults

Table 5 lists the principal settings used in the main experiments. Where multiple values are plausible, we adopt fixed defaults as controlled baselines rather than reporting an extensive tuning study; this choice should be interpreted as transparency about researcher degrees of freedom rather than a claim that each model is globally optimal.

Table 5: Key hyperparameters in the replication code (main run).

Component	Setting
$PM_{2.5}$ cap (main)	99.5th percentile of train sample
Lag features	Configurable; core autoregressive signal includes lag 24 (daily cycle)
Calendar	hour, day-of-week, month
Heating season	October–December and January–April indicator
Ridge	$\alpha = 1$, least-squares solver `lsqr`
Random forest	100 trees, max depth 10, `random_state=42`
XGBoost	100 estimators, max depth 6, $\eta = 0.1$, `random_state=42` (defaults; tuned values in JSON)
LightGBM	See repository defaults; Hyperopt can override (JSON)
AR benchmark	AR(24) with intercept, coefficients frozen after train fit
SARIMA (main run)	$(2, 1, 2) \times (1, 0, 0)_{24}$; IC-based order search on train only
Tree tuning (optional)	Hyperopt (TPE), MAE on validation, max evals in config

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References

American University of Armenia. Air pollution: sources, effects, and yerevan context. Capstone monograph, Computer Science, AUA, 2025. URL https://cse.aua.am/wp-content/uploads/2025/06/Air_pollution-1.pdf. Local Yerevan air quality overview; useful comparator for study design and public framing.

George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel, and Greta M. Ljung. *Time Series Analysis: Forecasting and Control*. John Wiley & Sons, 5 edition, 2015.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.

European Environment Agency. Air quality in europe — 2019 report. Technical report, EEA, 2019. Background on PM2.5 health and policy relevance in Europe.

David I. Harvey, Stephen J. Leybourne, and Paul Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997. doi: 10.1016/S0169-2070(96)00719-4.

Rob J. Hyndman and George Athanasopoulos. *Forecasting: Principles and Practice*. OTexts, Melbourne, Australia, 3 edition, 2021. URL https://otexts.com/fpp3/.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Massimiliano Marcellino, James H. Stock, and Mark W. Watson. A comparison of direct and iterated multistep AR methods for forecasting macroeconomic time series. *Journal of Econometrics*, 135(1-2):499–526, 2006. doi: 10.1016/j.jeconom.2005.07.020.

Xiaolong Shao, Xin Li, Xiaoyan Ma, and Yang Shi. Predictions of PM2.5 concentrations with machine learning techniques: A review. *Ecological Indicators*, 119:106899, 2020. doi: 10.1016/j.ecolind.2020.106899.

World Health Organization. Who global air quality guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. WHO guideline document, 2021. Summary guideline values for PM2.5 exposure.